

Feasibility study of using a biogas engine as backup in a decentralized hybrid (PV/wind/battery) power generation system – Case study Kenya



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ABSTRACT

In this study, a hybrid power system consisting of PV (Photovoltaics) panels, a wind turbine and a biogas engine is proposed to supply the electricity demand of a village in Kenya. The average and the peak load of the village are around 8 kW and 16.5 kW respectively.

The feasibility of using locally produced biogas to drive a backup engine in comparison to using a diesel engine as backup has been explored through a techno-economic analysis using HOMER (Hybrid Optimization Model for Electric Renewables). This hybrid system has also been compared with a single diesel based power system.

The results show that the hybrid system integrated with the biogas engine as backup can be a better solution than using a diesel engine as backup. The share of power generation by PV, wind and biogas are 49%, 19% and 32%, respectively. The LCOE (Levelized Cost of Electricity) of generated electricity by this hybrid system (\$0.25/kWh) is about 20% cheaper than that with a diesel engine as backup (\$0.31/kWh), while the capital cost and the total NPC (Net Present Cost) are about 30% and 18% lower, respectively.

Regarding CO₂ emissions, using a biogas engine as backup saves 17 tons of CO₂ per year compared to using the diesel engine as backup.

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1. Introduction

According to the IEA (International Energy Agency), roughly 1.5 billion people around the world do not have access to electricity [1,2]. Regarding water supply, the numbers provided by the WHO (World Health Organization) are also alarming: 1.1 billion people do not have access to any types of improved drinking source of water [3] and around 50% of them have some health issues related to lack of clean water sources and sanitation systems [4].

On the other hand, environmental impacts of providing electricity with the traditional energy sources, their limited supply and rising cost is a high concern for our modern society. Providing electricity for remote areas in emerging countries through the grid extension can be very expensive and usually needs a long term planning. A distributed power generation system (e.g. a micro-grid) which provides electricity from locally available renewable energy

sources such as solar energy, wind power and biomass might be a better solution for these countries. Therefore, the focus of this work is on providing electricity and clean water for emerging countries from locally available energy sources.

Kenya, with relatively high solar energy, has been chosen as a case study. Kenya is an Eastern African country with over 44 million people and according to the UN Human Development Index rank, it belongs to the low development countries. Besides other existing contributors to this current situation, one of the main reasons relies on the fact that Kenya is a country with low levels of electrification and basic infrastructure: according to IEA's information about modern energy access in developing countries, the Energy Development Index data, Kenya is in the 68th position out of 80 countries' ranking [2].

Kenya has a substantial proportion of unexploited renewable resources such as biomass, solar and wind [5]. However, its electrification rate is only 23% and, when assessing rural areas, only 8% have access to electricity [6]. In addition, when considering the grid extension option for Kenya, it can be extremely expensive. According to the report prepared by Alliance for Rural Electrification

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[7], an altogether cost of \$12,550 per kilometre (materials plus labour & other costs) is estimated. Concerning the water situation, and according to the Joint Monitoring Programme's 2012 report [8], access to drinkable water in Kenya is 59%.

In order to revert that situation and align it with the latest MDG (Millennium Development Goals), Kenya's government established a long-term strategy, the Vision 2030 [9]. Their objectives towards the energy goals are: to increase national power generation, provide the energy required to accelerate growth and mobilize private sector capital for generation of electricity from renewable energy.

At the moment providing electricity by diesel engine is one of the most common solutions for rural electrification [10]. Despite its simple installation and control schemes which are considered as advantages, the fuel cost, the transportation cost, bulk storage need and its environmental effects make it a very unappealing choice in the long term. Furthermore, the sharp rise of diesel engine price globally and in Kenya results in a steep rise of diesel engine operating cost [10].

A hybrid renewable micro-grid that provides electricity by solar energy and wind power seems to be a more sustainable solution. However, due to the intermittent nature of wind power and solar energy, the backup engine plays an important role in this system. In many rural areas, diesel engine has been used as backup. Another possibility is using locally produced biogas in a gas engine as a substitute for the diesel engine. This mitigates the draw backs of using diesel fuel; however, it might not be a feasible solution economically.

This study aims to understand under which circumstances using the locally produced biogas-driven engine as backup is more economic than using a diesel engine. Furthermore, the options of using only diesel engine and also the grid extension for rural electrification are briefly explored and the costs of generated electricity are compared.

Section 2 presents a literature review on the decentralized power system consists of battery, solar PV (Photovoltaics) or/and wind turbines using biogas or diesel engine as backup. In Section 3, objectives and methodology of the study have been discussed. Section 4 discusses the core characteristics of the study, i.e., its location and the energy supply and demand assumptions. Section 5 is dedicated to the results and their discussion. Finally, the conclusion section summarizes this study's findings.

2. Literature review

Electrification of small villages in rural areas is one of the first steps towards a better life quality in developing countries [10]. Access to electricity has a great influence on human health, economy and education. It also may prevent migratory flow to large cities and a society's disbelief regarding its own future [11].

One of the traditional solutions in remote and rural areas is to use a diesel engine which has a low upfront costs and it is relatively easy to install and control. However, the fuel price, transportation cost, bulk storage needs and environmental impacts of using the diesel engine make this solution less attractive. The new trend in rural electrification is extracting energy from locally available renewable energy sources in a hybrid configuration. Several works and research studies have been conducted in this field and most of them validate these types of hybrid energy solutions' advantages [7,12–18]. In many of them, solar and wind energy are considered as the primary energy sources but due to their intermittent nature, there is a need to have a backup system in case of shortage or lack of wind and solar energy [14]. In many cases diesel engine and battery bank are chosen as a backup generator and energy storage, respectively [1,19,20].

Numerous authors have studied hybrid systems consists of PV module, wind turbine, diesel engine and battery bank [7,11,13,21,22]. In many of them HOMER (Hybrid Optimization Model for Electric Renewables) has been used as a tool to analyse the system operation [11,15,18,21,23,24]. HOMER suggests a set of ranked optimized energy solutions, for which the criteria is the minimum NPC (Net Present Cost) and maximum power reliability.

Considering the literature, it is evident that a hybrid power system including a diesel engine has many advantages in comparison with a purely diesel-powered system, such as decreasing the fuel consumption and O&M (Operation and Maintenance) costs, increasing the lifetime of the engine and decreasing the negative environmental impacts. However, the problem with these kinds of systems is that they still rely on fossil fuel which is not environmentally friendly and requires logistics related to the fuel transportation to the community. Due to the high transportation costs and theft risk, local fuel availability is often low. To overcome the aforementioned problems, the diesel engine can be completely removed by substituting a biogas engine, the fuel of which can be provided locally in a small-scale digester. Several studies have considered biogas as a viable option to be included in an autonomous hybrid energy system [25–31].

The village presented in this work as a case study has the possibility of using animal manure as feedstock to the digester. Therefore, the focus of the following part is on the literature that have presented the option of using locally produced biogas from animal manure in small-scale digesters to produce electricity. Rahman et al. [30] proposed a hybrid energy system consists of solar PV arrays, a biogas digester and a biogas engine to provide the energy demand of households in Bangladesh. Biogas is produced locally from cattle manure in a small digester. The produced biogas is used both for cooking facility and electricity generation. The results show that if the household has more than three cattle, this system can provide the cooking and electricity demand of the household.

Pérez et al. [29] performed a comparative analysis between plastic tubular and fixed dome digesters for household application in rural communities. It has been concluded that the plastic tubular digesters have lower investment cost and they are easy to install and maintain compared to the fixed dome digesters that have lower environmental impacts.

Garfi et al. [27] evaluated the technical, environmental, economic and social impacts of using small-scale low-cost household digesters in rural communities. Although in the pilot project most of the yielded biogas has been used for cooking purposes, it has the potential to be used for electricity generation and in some cases it has already been used for lighting purposes.

Borges Neto et al. [26] explored decentralized energy supply of rural areas by a hybrid system consists of solar photovoltaic and biogas digester. It is observed that in addition to providing energy, this system can create job and improve the economy of the household within the community. Furthermore, it is mentioned that by sufficient technical and economic support of the government, the system can be taken into consideration as a Clean Development Mechanism.

In Refs. [25,31], the feasibility of providing electricity by a small-scale hybrid energy system consist of solar PV/wind/biogas/diesel in remote areas in India and Bangladesh has been explored in two different studies. HOMER has been used as a tool to perform the optimization and to find the most feasible solution in both studies. The results show that using locally produced biogas to drive a backup engine is a viable solution for remote and isolated area and the integrated system has higher feasibility compared to the grid extension.

An overview of the common biogas production systems in Kenya was made by Nzila et al. [28]. A comparative analysis considering techno-economic and environmental aspects of these systems has been performed in a LCSA (Life Cycle Sustainability Assessment) framework. Based on this assessment, in case of using animal manure as the main feedstock, the tubular and the fixed dome digesters are examined to be the most sustainable option.

3. Objectives and methodology

As it is mentioned in literature review section, small-scale household digester to produce biogas from manure feedstock has been investigated in several studies. However the possibility of being used in a backup engine as a substitute for diesel engine in a hybrid system consists of both solar PV panels and a wind turbine is not explored thoroughly enough.

Using locally produced biogas from manure can eliminate problems related to diesel fuel use. However, including a digester adds additional expenses to the system such as the cost of the digester system, collecting manure and feeding the digester on a daily basis.

Therefore, the main purpose of this study is to assess biogas engine viability when being used in a hybrid system as backup compared to the one using diesel engine. It is also investigated under which circumstances the use of locally produced biogas is more feasible. Since using only diesel fuel is one of the most common solutions for power generation in rural areas, the economics of this system is needed to be analysed and compared with the hybrid one. Therefore, a techno-economic analysis of a diesel-powered system has been performed additionally. Furthermore, since grid extension is another option for rural electrification, it has been investigated in brief.

HOMER software has been used as a tool to find the group of optimized systems that fulfil the load demand under a particular input assumptions and system constraints [24]. HOMER is an

optimization tool for distributed power, developed by National Renewable Energy Laboratory, USA.

Due to the uncertainties, a sensitivity analysis has been performed to determine the effect of input assumptions on the optimization result. The methodology and general approach of the system modelling and analysis is shown in Fig. 1.

4. System modelling

A model consists of a wind turbine, solar PV panels, battery bank and diesel/biogas engine has been implemented in HOMER. In the following section, the model input and assumptions such as meteorological data and energy resources for the selected location, load demand of the community, sensitivity variables and components' parameters are described. The optimization and sensitivity analysis results are depicted later in the next section.

4.1. Garissa district

Garissa is a district located in North Eastern province in Kenya. Due to its low elevation and being located far from the coast, it has a very warm climate and the landscape is mostly arid or semi-arid. The rate of poverty in Garissa is about 49% [32] which is more than Kenya's average value. The total population of Garissa county and households distribution in Garissa district are shown in Tables 1 and 2, respectively. As it can be seen, more than 75% of the county's population lives in rural area and in Garissa district, in which the study has been performed, over 11,000 households are located in the rural areas.

4.2. Primary energy sources

In this section, the primary energy sources and their availability are discussed briefly. Solar insolation and wind data are extracted

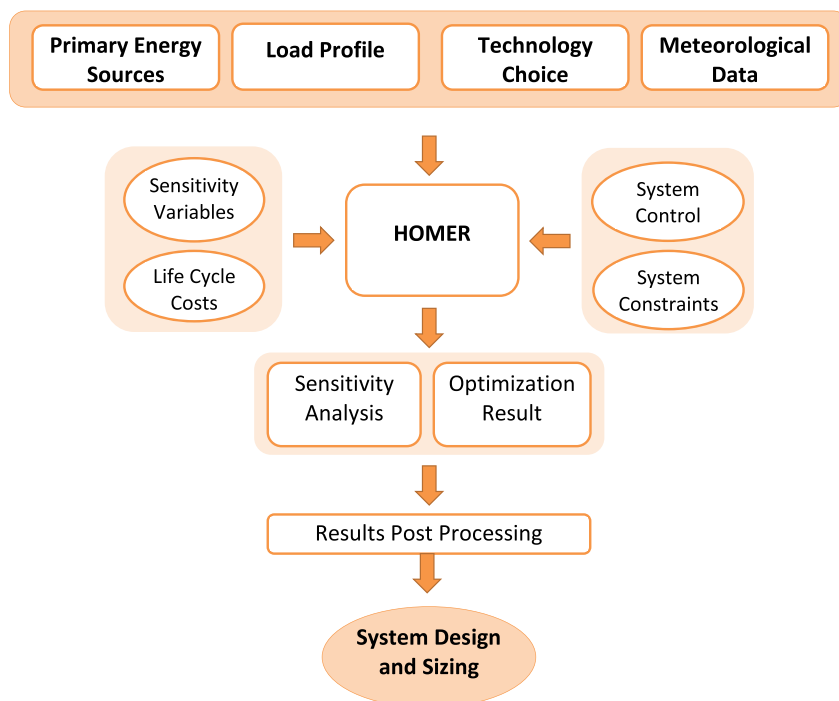


Fig. 1. Methodology and general approach of system analysis.

Table 1

Garissa County and Kenya's average rural distribution data [32].

County	Rural	Urban	Total	% Rural	% Urban
Garissa	476,392	146,668	623,060	76.5%	23.5%
Kenya average	26,122,722	12,487,375	38,610,097	67.7%	32.3%

Table 2

Garissa district – households distribution [32].

District	Rural/urban	Total	Households	Population density
Garissa	Urban	116,953	20,929	56,572
Garissa	Rural	73,109	11,189	32,384

from the Transient System Simulation program (TRNSYS 17) using the Meteororm data base [33].

4.2.1. Solar energy

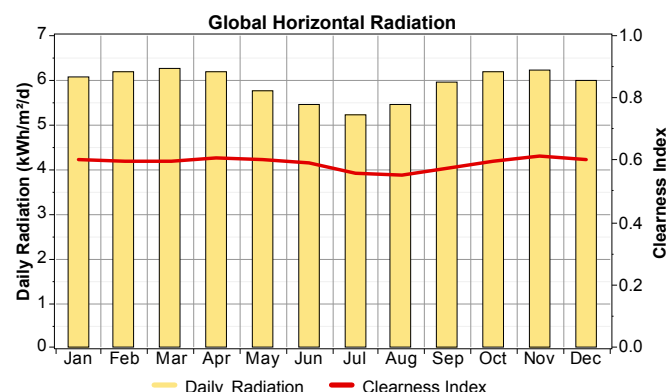
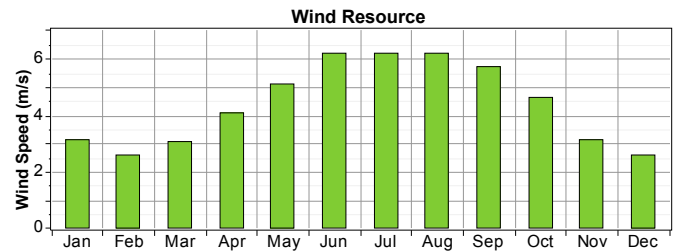
Due to the geographical location of Kenya, solar energy potential is relatively high and many studies have emphasized the importance of using solar energy as one of the most available primary energy sources in Kenya. The average amount of solar power is about 5 kWh/m²/day but despite this fact the share of solar energy in Kenya power generation is very low [5]. According to the Meteororm and data extracted from the TRNSYS tool [33] the average amount of solar power in Garissa is 5.9 kWh/m²/day. The monthly average global horizontal solar radiation of this city is shown in Fig. 2.

4.2.2. Wind power

Considering the whole country, Kenya does not have strong winds to drive large scale wind turbines except in certain parts of Kenya such as Turkana, Marsabit, Ngon or the coastal region with average wind speed ranging from 8 to 14 m/s [5,34]. However, the average wind speed in many locations is capable of driving small wind turbines [5,34]. According to the Meteororm data base, the annual average wind speed in Garissa is 4.4 m/s. The Monthly average wind speed in Garissa during a year is shown in Fig. 3.

4.2.3. Biogas

Fifty percent of the biomass in Kenya are supplied by farm land in different forms such as wood biomass, crop and animal wastes [5]. The focus of this paper is only on producing biogas from animal manure. Different investigations in Africa have shown that there is a great potential for biogas production from animal manure in Kenya [35].

**Fig. 2.** Global horizontal solar radiation – Garissa, Kenya.**Fig. 3.** Wind resource – Garissa, Kenya.

Data taken from Kenya Open Data [32] shows that the number of cattle and sheep in the Garissa district are around 260,000 and 310,000, respectively (Table 1). These numbers indicate that there is a high potential for biogas production from cattle and sheep dung.

Since collecting the cattle manure is easier than goats and sheep in a dispersed area, in this study cattle manure is assumed to be the main source for biogas production and sheep manure is the second choice in case of shortage of cattle manure. It is important to mention that the methane content in yielded biogas from sheep and cattle manure is 70–72% and 56–60%, respectively [36]. The average biogas production from cattle manure is assumed to be 0.96 m³/cow/day [37]. Regarding data in Tables 2 and 3 and considering the limited number of cattle in the village, biogas production can be predicted.

Since the average biogas production from cattle can change due to the cattle race, their nutrition diet and the size of cow, the biogas production by each cow is taken to be 0.8 m³/cow/day. By assuming 150 cows in a village of 100 family houses, the biogas production is about 120 m³/day and being conservative the biogas production is taken to be 50% of this amount which is 60 m³/day and around 20,000 m³/year. This assumption is used as one of the constraints in HOMER model.

4.2.4. Diesel

As it has been mentioned before, providing electricity by single diesel engine is one of the most common solutions for rural electrification [38]. Despite its simple installation and control schemes which are advantages, the fuel cost, the transportation cost, bulk storage need, high O&M cost of the engine and its environmental effects make it a very unappealing choice in the long term. Furthermore the sharp rise of diesel engine price globally and in Kenya results in a steep rise of diesel engine operating cost [38]. Diesel fuel needs to be transported to the village.

4.3. Electricity demand

The aim of this hybrid system is to provide electricity and clean water for 100 households, with an average size of 5 people. Each house is assumed to have a limited number of efficient lamps, a fan, a radio, a mobile charger, a small TV, a small refrigerator and a RO (reverse osmosis) water purification system. The load demand details and daily load profile are respectively shown in Table 4 and Fig. 4. The assumptions regarding considered appliances, their daily usage and their power consumption were taken from different sources in literature [1,39–42]. As it can be seen in Table 4, an extra

Table 3

The number of cattle, sheep and goats in Garissa district [32].

District	Cattle	Sheep	Goats	Camels	Donkeys
Garissa	266,900	312,600	1,000,900	101,500	25,800

Table 4
Daily electrical load.

Appliance	Electricity demand per household (Wh/Day)	Total electricity demand for 100 households (kWh/Day)
Light	160	16
Radio	200	20
Mobile charging	30	3
TV	325	33
Fan	275	28
Small refrigerator	720	72
Water purification system	—	4
Other appliances	180	18
Total	1890	190–200

power (other appliances) is assumed in case of using any other equipment, such as a laptop or other non-permanent appliance. Unlike other authors [11,43], this study only focused on the household's electricity demand.

To purify local water, a water purification system is added to the system. By assuming an average water demand (4 L/person per day [44]), 2 m³ of purified water per day are required. Since the power consumption of the RO system is very low, this technology has been chosen as the water purification system. In this study a reverse osmosis water purification system (TW-1.2 K – 225) from Pure Aqua, Inc. has been chosen as a case reference [45]. The average load and the peak load during the day are around 8 kW and 16.5 kW, respectively.

The profile shown in Fig. 4 matches the typical electricity load demand in rural areas suggested by IEA [1], i.e., a lower demand level during night hours, medium demand during morning and mid-day and an evening peak load.

4.4. The hybrid system configuration

The stand-alone hybrid power generation system consists of PV panels, wind turbine, biogas/diesel engine and battery storage. A schematic of the system configuration is shown in Fig. 5 PV panels, wind turbine and battery bank are connected to a DC-Bus and the diesel/gas engine generator set is connected to an AC-Bus. These two buses are connected to each other using a bi-directional inverter. Load demand can be AC or DC, however, in the model presented in this paper, the load is assumed to be AC.

4.5. Component specifications

In the following section, the components' specification and their costs are described. It is worth to mention that finding the true costs of these components is hard and it can be significantly different from one market to another and also can change frequently. However a report published by IRENA (International

Renewable Energy Agency), “renewable power generation costs in 2014”, is used as a reference in order to reduce uncertainty about the costs of renewable power generation technology [46]. In this paper, in addition to the market survey and available literature, this report has been used as a check point for a more accurate cost estimation.

4.5.1. PV

PV Panels with the lifetime of 25 years are connected to a DC output. There is no tracking system and the slope and azimuth angles are assumed to be the average optimum during the years which are 0° and 159°, respectively. These optimum angles have been calculated by an online calculator in PVGIS (Photovoltaic Geographical Information System) website [47].

The ground reflectance and de-rating factor are assumed to be 20% and 90%, respectively. The intent of using the de-rating factor is to account for different types of losses in the PV panels such as shading, aging, wiring losses and soiling in the real world operating condition [24]. The capital cost and replacement cost of the PV panels are assumed to be \$2800/kW and \$2500/kW, respectively. Installation and shipping are including in these prices. The O&M cost of the PV panels is relatively low and is assumed to be \$10/kW/year.

4.5.2. Battery

Due to the intermittent nature of renewable energy, energy storage should be considered in the system design. In this model battery bank storage is chosen to maintain the constant voltage during the peak load or in the occurrence of a shortfall. A 6 V battery (Surrette 6CS25P) with a nominal capacity of 1156 Ah and with the lifetime throughput of 9645 kWh is selected from the HOMER data base. Since the voltage of the DC bus is assumed to be 24 V, four batteries per string are needed. The capital cost and replacement cost of the battery are \$1200 and \$1,000, respectively and \$15/year is assumed for the O&M of the battery. Since there is an engine generator set as a backup in the system, the number of autonomy hours is design to be up to 24 h.

4.5.3. Converter

Since there are both a DC bus and an AC bus in the system there is a need to have an inverter and a rectifier. In HOMER, a converter (that has both the function of a rectifier and an inverter) does the power conditioning in the system. The lifetime of the converter is assumed to be 20 years and the efficiency of the inverter and rectifier are both taken as 90%. The capital and replacement cost of the converter are assumed to be \$750/kW and \$700/kW, respectively. The O&M cost is negligible and is assumed to be zero.

4.5.4. Wind turbine

A DC generic 10 kW horizontal axis wind turbine with the hub height of 25 m and the lifetime of 25 years is selected from the HOMER data base. The capital cost of the wind turbine for one unit is taken as \$20,000 with the replacement cost of \$15,000 and O&M cost of \$200/year.

4.5.5. Diesel engine generator set

The capital cost, the replacement cost and the O&M cost of the diesel engine are assumed to be \$600/kW, \$500/kW and \$0.02/kWh, respectively. The minimum load ratio is taken to be 40% of the engine capacity and the generator operating hours are taken as 20,000.

4.5.6. Biogas engine and digester

One of the purposes of this study is to perform a comparative techno-economic analysis between diesel engine and biogas engine

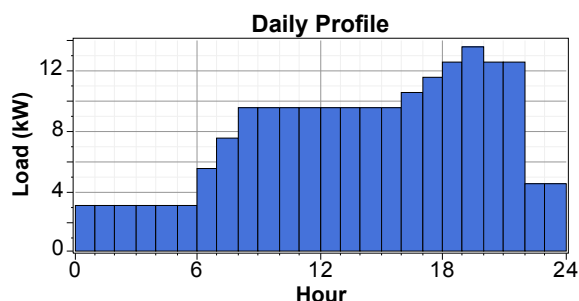


Fig. 4. Daily load profile.

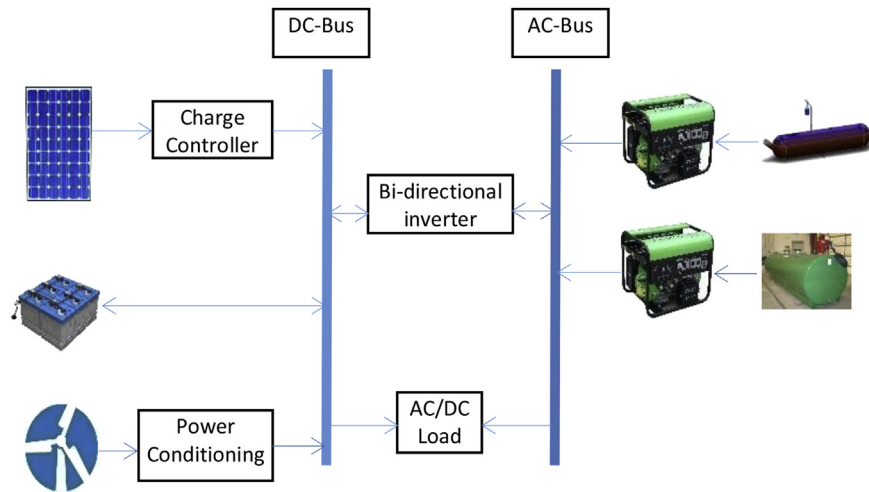


Fig. 5. Schematic of hybrid system configuration.

operation. The price of gas engine and diesel engine are slightly different in the market and the fuel price might influence the O&M cost of the system. The biogas is produced locally from cattle manure in the community and this biogas production can be relatively expensive. There are many commercial biogas digesters in the market that are very different in operation, construction, the amount of biogas production and cost. To make the biogas engine more cost effective, cheap solutions which are available locally should be considered in the system design. In this case study, a plug flow reactor which is a dry anaerobic digestion technology has been selected [48]. The reactor is chosen to be an in-ground tubular tank which is relatively cheap with easy installation and maintenance scheme. The total solid percentage of the input material is around 20–40% and the HRT (Hydraulic Retention Time) is around 20–30 days.

By doing a market survey on the capital cost of the available biogas engine with digester system including piping and cleaning component, a price of US \$1500/kW and US \$1200/kW are taken as the capital cost and the replacement cost, respectively.

Both the engine and the digestion system operation costs are included in the O&M of the system. Feeding the digester with substrate on a daily basis is also included in this price. Since the biogas is produced locally, the fuel price is included in the O&M cost. The O&M cost is dependent on the price of daily labour work for collecting the substrate, preparation and feeding to the digester. The digester system also should be monitored on a daily basis.

Considering the average salary in Kenya [49] and assuming that to produce 60 m³ of biogas, 2 h of labour work per day are needed to keep the system running, the O&M of the digestion system will be around \$0.08/kWh. Assuming the engine O&M to be \$0.02/kWh, its total O&M will be \$0.1/kWh. Since this price is based on a rough assumption, a sensitivity analysis has been done to have a better projection of the system operation.

To summarize all the data described in this section, an outline of the system's components characteristics as HOMER input data is

presented in Table 5. It presents a summary of the system's components characteristics, as HOMER input data.

4.6. Optimization criteria and sensitivity variables

In order to define the group of optimized system, HOMER performs simulation and optimization by doing an energy balance for each hour of the year considering the optimization criteria. The optimization objective is to determine the decision variables (size or numbers of each component) of a system with the minimum cost that is capable to satisfy the energy demand within the imposed constraints while maximizing the power reliability. In HOMER, the power reliability is defined as a parameter called capacity shortage that is the amount of power shortage divided by the total power demand during one year operation. The optimization is constrained to the maximum allowable capacity shortage defined by the designer.

The economy of the system and the optimization result are significantly dependent on the diesel fuel price, the capital and O&M cost of the biogas engine generator set system and the maximum allowable capacity shortage. To understand the effects of aforementioned parameters, a sensitivity analysis has been performed. The decision variables taken into account in the optimization process are the number of PV modules, wind turbine size, battery bank capacity, size of the engine and size of the inverter. The sensitivity variables are shown in Table 6. The costs related to the biogas engine are defined as a multiplier relative to the base cost assumption (Table 5) which is explained more in the sensitivity results' section thoroughly.

4.7. Other parameters

The lifetime of the project is taken as 20 years and the annual real interest rate is assumed to be 6%. In the simulation process these values are applied to each component in order to calculate

Table 5
Summary of HOMER input.

Component	Size	Capital cost (\$)	Replacement cost (\$)	O&M cost	Quantities to consider
DC wind turbine	10 kW	20,000	15,000	200 (\$/yr)	0,1 (Units)
Diesel engine Gen-Set	1 kW	600	500	0.02 (\$/hr)	0,5,10,15 (kW)
Battery	1156 Ah	1200	1000	15 (\$/yr)	0, 8, 16, 24, 32 (Units)
Converter	1 kW	750	700	0	0, 10, 15 kW
PV	1 kW	2800	2500	10 (\$/yr)	0, 20, 25, 30, 35, 40 (kW)
Biogas engine Gen-Set with digester	1 kW	1500	1200	0.1 (\$/hr)	0,5,10,15 (kW)

Table 6
Sensitivity variables.

Sensitivity variable	Value
Diesel price (\$/L)	0.5, 0.7, 0.9, 1.1, 1.3, 1.5
Capacity shortage (%)	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Biogas Gen Set	Capital cost multiplier
with digester	Replacement cost multiplier
system	O&M cost multiplier
	0.5, 1, 1.5, 2, 2.5, 3

the system NPC. In the model, the load following dispatch strategy has been chosen and the annual capacity shortage is assumed to be 3%. To determine the influence of the capacity shortage on the LCOE (Levelized Cost of Electricity) and the total NPC, a sensitivity analysis has been performed assuming the capacity shortage percentage to be between 0 and 10 (Table 6).

5. Results and discussion

In this section, the results of the model are presented. In the first part, the optimization results are analysed and after that, the sensitivity analyses are described.

5.1. Optimization results

Similar to the other optimization tools, HOMER results show a group of optimized systems. The optimization result is based on the diesel price of 1.1 \$/L, the biogas engine capital cost of \$1500 and the biogas engine-digester replacement cost of \$1200 and the O&M cost of \$0.1/kWh. The top four optimized systems are shown in Table 7. The NPC is considered one of the most important decision making parameters. As it is shown in this table, system number one has the lowest NPC compared to the other systems.

Table 7
Optimization results.

Group of optimized systems				Only diesel + battery		
System configuration	Unit	1	2	3	4	
PV panel	kW	20	30	35	30	0
10 kW wind turbine	Number	1	0	1	0	0
Diesel Gen Set	kW	0	0	5	10	15
Biogas Gen Set	kW	10	10	0	0	0
Surrette 6CS25P	Number	16	24	32	32	8
Converter	kW	10	10	10	10	15
Economic	Total capital cost	\$ 117,700	135,300	166,900	135,900	29,850
	Total NPC	\$ 196,700	224,800	241,000	299,800	445,830
	Tot. ann. cap. cost	\$/yr 10,262	11,800	14,550	11,850	2600
	Tot. ann. repl. cost	\$/yr 2100	2580	765	1570	2950
	Total O&M cost	\$/yr 4800	5225	1310	1680	2530
	Total fuel cost	\$/yr		4400	11,030	30,790
	Total ann. cost	\$/yr 17,145	19,600	21,010	26,140	38,870
	LCOE	\$/kWh 0.254	0.289	0.312	0.387	0.56
Power production	PV panel	kWh/yr 38,990	58,340	68,065	58,340	0
	Wind turbine	kWh/yr 15,030	0	15,030	0	0
	Diesel Gen Set	kWh/yr 0	0	10,220	22,955	78,225
	Biogas Gen Set	kWh/yr 25,080	26,155	0	0	0
	Tot. electrical production	kWh/yr 79,000	84,500	93,315	81,300	78,225
	Ren. fraction (%)	100	100	89	72	0
	Cap. shortage	kWh/yr 1945	1675	1940	1800	0
	Unmet load frac.	0.03	0.02	0.03	0.03	0.00325
	Excess electricity	kWh/yr 7120	11,260	20,805	7800	651
Diesel Gen Set	Diesel Gen Set fuel	L/yr 0	0	3980	10,030	27,990
	Diesel Gen Set hours	hr/yr 0	0	2800	4500	8030
	Diesel Gen Set Starts	starts/yr 0	0	290	444	730
	Diesel Gen Set life	Yr 0	0	7.15	4.45	2.5
Biogas Gen Set	BiGen Gen Set fuel	m3/yr 16,265	17,250	0	0	—
	BiGen Gen Set hours	hr/yr 4150	4565	0	0	—
	BiGen Gen Set starts	starts/yr 610	495	0	0	—
	BiGen Gen Set life	Yr 4.8	4.4	0	0	—
Battery	Battery autonomy	Hr 8	13	17	17	4
	Battery throughput	kWh/yr 9100	13,600	18,800	16,990	3180

However, a cheaper system is not necessarily a better solution and the best choice should be determined in the post processing stage. This selection is always a trade-off between the system cost and other criteria which are important for the designer. Other parameters which might be important to find the best system are the engine operating hours, the biogas consumption, the diesel fuel consumption and the battery autonomy.

Due to the environmental issues and other terms stated earlier, using the diesel engine generator set should be avoided as much as possible. It means that system number three and four with diesel generator set as a backup system have lower priority than the first and the second configurations. System number one has lower biogas consumption and a smaller battery bank compared to the system number two. The variety of using different types of local energy sources might be considered as an advantage for a hybrid system. In system number one, energy is extracted from three types of energy sources (Wind, Solar and Biogas) and has priority to the system number two in which energy is extracted from two types of local energy sources (Solar and biogas).

Bearing in mind the above mentioned criteria, system number one is considered to be the best optimized system in this case study. This system is composed of a 20 kW PV-Array, 16 Surrette 6CS25P batteries, a 10 kW wind turbine, a 10 kW biogas engine generator set and a 10 kW inverter. The amount of generated power and the share of power generation by each component for this system during one year operation are shown in Table 8, and the monthly average electricity production by each component is presented in Fig. 6.

In this configuration, the PV array dominates the power generation by providing 49% of the total generated power. The monthly average share of power generation by solar panel fluctuates slightly during the year and it is always remains between 46% and 56%. On

Table 8

Share of power generation by each component during one year operation.

Production	kWh/yr	%
PV array	38,900	49
Wind turbine	15,030	19
Biogas Gen-set	25,080	32
Total	79,000	100

the other hand, the monthly average share of wind power generation changes dramatically during the year (from 5 to 38%) and has the maximum during the summer and minimum value during the winter. To compensate the shortage of wind or solar power, the biogas engine generator set delivers the remaining power demand. The monthly average share of generated power by biogas engine generator set changes between 23% and 42% throughout the year. The capacity factors of PV array, wind turbine and engine generator set are 22.3%, 17.2% and 28.5%, respectively.

The total capital cost, the total NPC and the LCOE of this system are \$117,700, \$196,389 and \$0.254/kWh, respectively. This system also has the lowest initial capital cost which is an advantage. The cash flow summary of the system by cost type which is shown in Fig. 7 shows that almost 50% of the capital cost goes to the PV system and the capital cost of the wind turbine and the battery bank are almost the same.

As been cited in the preceding section, the price of the biogas fuel as an input to HOMER is assumed to be zero and as a consequence the fuel cost in the cash flow has resulted in zero. Since the fuel is produced locally, the cost of fuel production is included in the biogas production process which is the operating cost of the biogas engine generator set in the model. That is why the O&M cost of the biogas generator set is quite high compared to the other components.

This figure also shows that only two components (Battery and Biogas Generator Set) are needed to be replaced during the project lifetime. As it is shown in the figure, around 75% of the total replacement cost goes to the biogas generator set and the remaining is allocated to the battery bank.

The cash flow summary of the system by each component is shown in Fig. 8. It is evident that the operating and the replacement cost of the biogas engine generator set is higher than the capital cost which is in contrast with the other components in which the most investment goes to the capital cost.

The unmet load and the excess electricity in this configuration are 2020 kWh/yr and 7200 kWh/yr (2.91% and 9.11% of the total electrical production), respectively. The unmet load fraction is a user defined parameter and is assumed to be maximum 3% in this

case. The excess electricity has to be dumped in this system and it might be considered as a negative point. However, it also shows the capability of the system to meet the future demand growth.

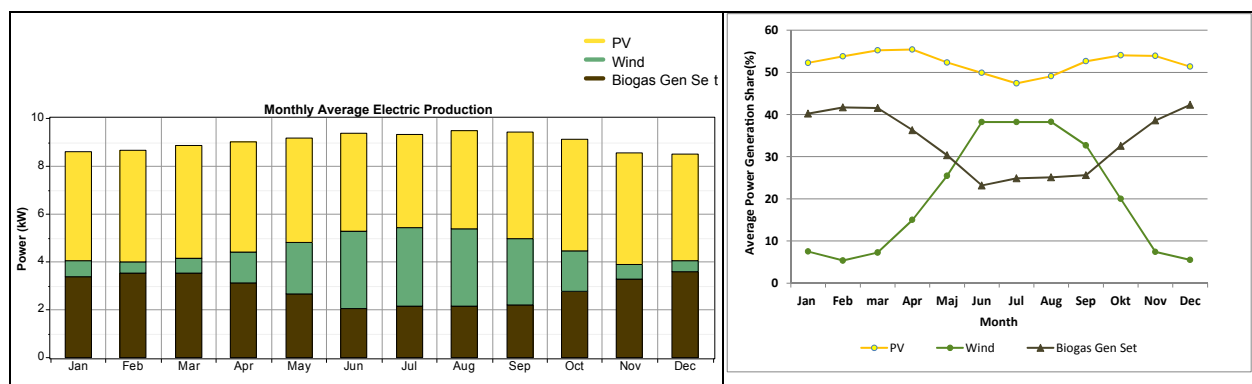
In order to have a better understanding of the system operation, details of the system's power flow and the battery status during 72 h are shown in Figs. 9 and 10, respectively. The demand load, the unmet load, the excess electricity and the amount of generated power by each component in March 20, 21 and 22 are shown in Fig. 9. As it is evident from the figure, the excess power mainly happens due to the high solar insolation during the day (10:00–16:00) and the unmet load mainly occurs during the evening between 19:00 to 21 due to the lack of solar insolation. Allowing the system to have electricity shortage to some extent results in lower NPC, however, the importance of power availability should always be examined. In this case since the system provides electricity for residential purposes in rural areas, 3% annual capacity shortage is a reasonable assumption. To overcome this shortage, consumers can have a schedule to avoid the usage of some of the appliances in that period of time.

The battery input power and the battery SOC (state of charge) for these 72 h are shown in Fig. 10. Even though there is some amount of excess electricity, the SOC of the battery remains mostly less than 70% and the battery discharges before it reaches the maximum storage capacity. Since there is an engine backup in the system, the autonomy of the battery is decided to be up to 24 h and the maximum number of batteries in the battery bank is limited to be 32 in the model.

The model results show that the battery input and output are 10,085 kWh/yr and 8120 kWh/yr, respectively. The difference between the battery input and output is mainly due to the charging and discharging losses (1890 kWh/yr) and storage depletion (67 kWh/yr). Expected battery life-time is calculated to be 12 years.

As noted earlier, using a diesel engine is a common solution to provide electricity in remote areas. This solution has been modelled with HOMER and the results are shown in Table 7. As it can be seen, the initial capital cost of the system is relatively low but the total NPC is quite high. The total capital cost, the total NPC and the LCOE of this system are \$29,850, \$445,830 and \$0.56/kWh, respectively.

A comparison between the system number one (the optimized hybrid solution) and the diesel based power generation shows that the NPC and LCOE are both around 55% lower in system number one. This is mainly due to the fuel cost and about 75% of the NPC goes to the diesel fuel during the projected life-time of the system. Another important parameter in Table 7 is the engine operating hours which is directly related to the maintenance cost of the system. Operating hours of the diesel engine (8000 h) in the single diesel engine based power generation system is double compared

**Fig. 6.** Monthly average electricity production.

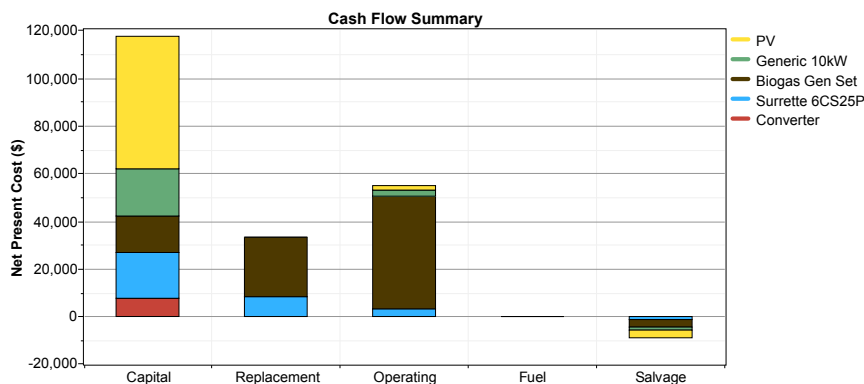


Fig. 7. Cash flow summary of the optimized system by cost type.

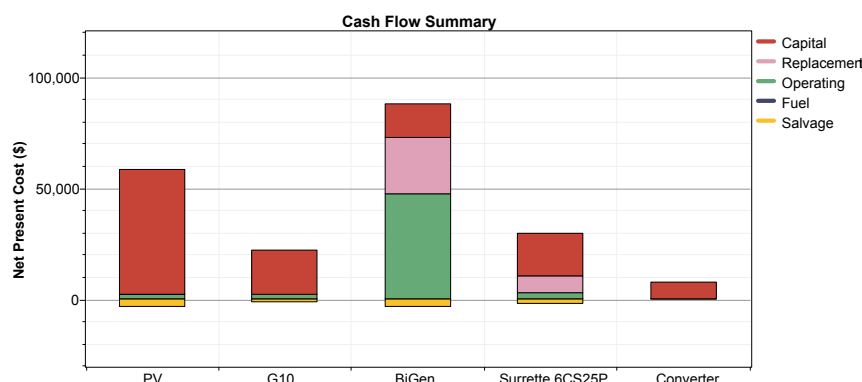


Fig. 8. Cash flow summary of the optimized system by component.

to the gas engine operating hours (4000 h) in the hybrid configuration.

Grid extension is another option for rural electrification in Kenya. As of this writing, a fixed monthly cost of \$1.59 and consumption tariff of \$0.03/kWh for the first 50 kWh, \$1.36 for consumption between 50 and 1500 kWh are charged by KPLC (Kenya Power and Lighting Company) [50,51]. There is also an additional upfront connection fee of \$825, which is an obstacle for low-income potential customers in rural areas [51].

Considering the above mentioned costs, customers in rural areas will pay roughly ten times more if the electricity is provided by a hybrid renewable power system such as the one proposed in this paper. In some literature this number is predicted to be twenty times more in a micro-grid [51]. However, grid extension needs a long term planning, proper infrastructure and expertise. Besides that, it is expensive and needs a long term investment. In the report prepared by Alliance for Rural Electrification [7] the cost of grid extension including the materials, labour and other costs is stated to be \$12,550 per kilometre. Concerning this number and also considering the optimized system as a reference the break-even point of the grid extension is 3 km which is a relatively short distance. It is evident that the proposed stand-alone system is more viable for remote and rural areas with low-consumption consumers.

5.2. Sensitivity results

As stated previously, the optimization results are based on the verified prices and the estimated prices taken from different websites and contacting with several companies [52]. Due to the

uncertainties related to the cost assumptions, a sensitivity analysis has been done to attain a more realistic system design.

The capital cost and the O&M cost of the biogas engine generator set varies dramatically in the market. This is due to the difference in the prices offered by manufacturer and also the cost of the O&M of the engine and the digester system. Due to the transportation cost and market fluctuations, the diesel price also can vary significantly locally and from time to time [53].

The sensitivity results for the three chosen parameters are shown in Fig. 11. The optimization results, which are described in the previous section, are based on the values mentioned in Table 5 (the diesel price of 1.1 \$/L, the biogas system capital cost of \$1500

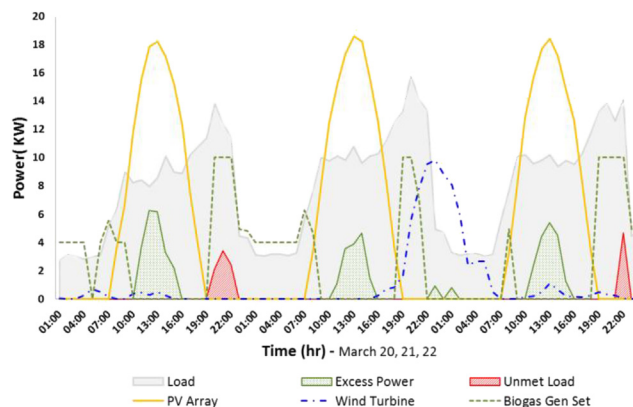


Fig. 9. Demand load, unmet load, excess power and generated power by each component during 72 h (March 20, 21, 22).

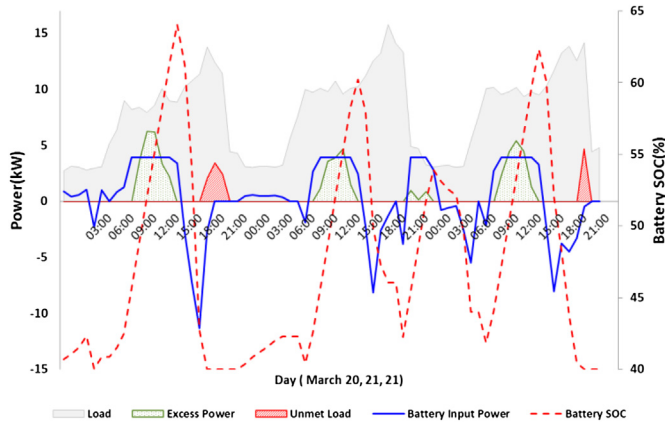


Fig. 10. Demand load, excess power, unmet load, battery input power and battery state of charge for 72 h (March 20–22).

and replacement cost of \$1200 and the O&M cost of \$0.1/kWh). Based on these assumptions, the backup system for the optimal hybrid configuration is the biogas engine. By changing each of these assumed costs, the backup system for the optimal design might change to diesel engine and the total NPC (the numbers in the figure) might be different.

As it is shown in Fig. 11, the biogas engine related costs are defined as a multiplier relative to the base assumptions. For example, a capital cost multiplier of 1 means that the capital cost is equal to the cost mentioned in Table 5. Furthermore, the capital cost multiplier of 0.5 and 1.5 means that the capital cost is fifty percent less and fifty percent more than the cost mentioned in Table 5, respectively.

To be more clear, looking at Fig. 11, if the diesel fuel price decreases to (\$0.8/L), O&M costs remain the same as basic assumption (one) and the capital cost of the biogas system becomes triple, a system with the diesel engine as backup is economically a better choice (Fig. 11). Similar graphs can be extracted for different O&M costs.

The surface plot of cost of electricity for a fixed diesel price of \$1.1/L is shown in Fig. 12. This figure shows how the LCOE is related to the capital cost and O&M cost of the biogas system.

Another design parameter which might change the optimized configuration is the maximum allowable capacity shortage. Power availability is one of the most important parameters in power system design but it should be always a trade-off between the cost and the consequence of power shortage. For example in a hospital due to the high importance of the power availability the capacity shortage should be zero but in this case study, since the system provides electricity for the residential sector, an annual capacity shortage of maximum 3% is assumed to be allowed in the system. In Fig. 13, influence of the maximum annual capacity shortage on the total NPC and the LCOE of the optimized configuration is shown. In this case, by increasing the annual allowable capacity shortage from 0% to 10%, the NPC and the LCOE decreases about 22% and 14%, respectively.

5.3. Carbon emissions analysis

Based on the LCA (life cycle analysis) available in literature, the equivalent CO₂e (CO₂ emission) for power generated from biogas and diesel fuel is estimated to be 11 gCO₂e/kWh and 680 gCO₂e/kWh (Table 9). It means that using biogas engine as a substitute for diesel engine save 670 gCO₂e/kWh which is a significant advantage.

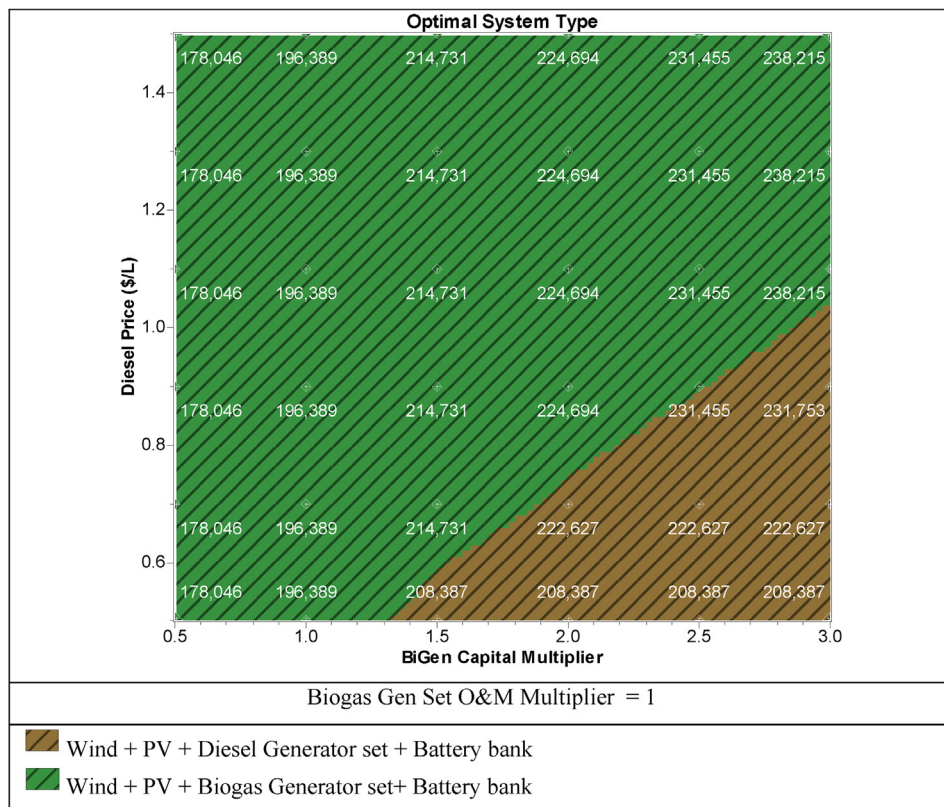


Fig. 11. Sensitivity analysis in term of diesel fuel price; Biogas engine generator set O&M and capital cost.

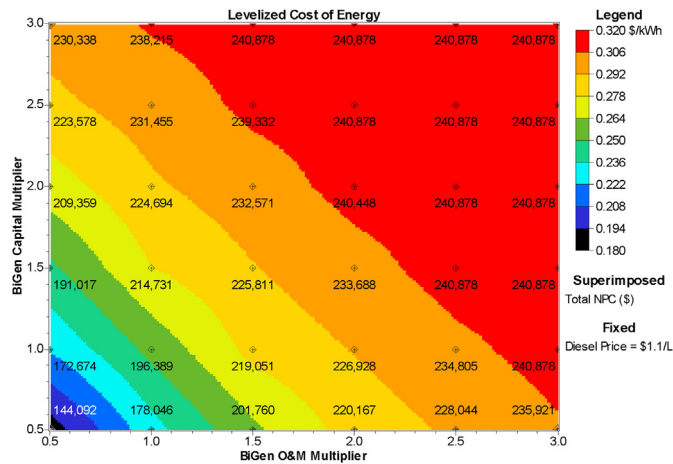


Fig. 12. Surface Plot of cost of electricity.

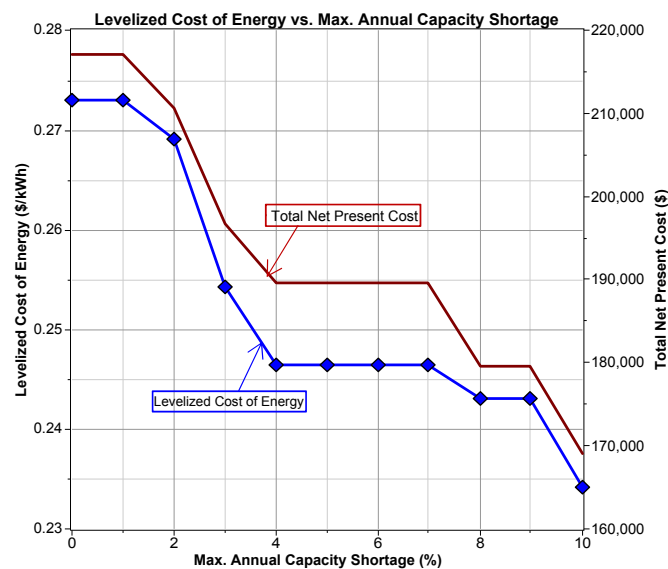


Fig. 13. Line graph for total NPC.

The total power and the generated power by biogas engine are 75000 kWh/year and 25000 kWh/year in the optimized hybrid system. Therefore, using this system saves around 17 tCO₂/year and 48 tCO₂/year in comparison with using the diesel engine as backup in the hybrid scheme and using only diesel engine, respectively.

The equivalent carbon emissions of other components in the system are shown in Table 9. Since different studies use different methods and parameters while doing the life cycle analysis, the value of emission factor might be different from one to another. Furthermore each technology has its own characteristics and efficiency which might change the value of the emission factors. Based on a rough estimation, using hybrid system with the biogas and diesel engine as backup saves around 22 tons of CO₂ emissions and 5 tons of CO₂ emissions per year, respectively, in comparison with the current grid in Kenya.

Table 9
Life cycle emission factor for power generators grid emission Kenya.

Power generator		Wind [54,55]	Solar PV [55,56]	Biogas [55]	Diesel [57]	Grid emission in Kenya [58]
CO ₂ emissions	Unit	1 kWh _{el}	1 kWh _{el}	1 kWh _{el}	1 kWh _{el}	kWh _{el}
	g	10	30–100	11	680–780	332

6. Conclusion

A techno-economic feasibility study of a decentralized power generation system for a rural area in Garissa district in Kenya has been performed. The generated electricity provides the daily power demand of a village with 100 family houses. This hybrid system is composed of PV arrays, a wind turbine, battery banks as storage and a biogas engine as backup system in which the biogas is produced locally by feeding a digester with cattle manure.

Since using diesel engine for power generation is a common solution in rural areas, a comparative analysis between a system with biogas engine and diesel engine as backup has been performed using HOMER software. The optimization result determines the group of optimized systems according to the cost assumptions and techno-economic constraints. Based on the optimization results, the hybrid system with biogas engine as backup is appeared to be the best option. In this system, the share of renewable energy in power generation is 100%.

The LCOE of generated electricity by this hybrid energy system is calculated to be \$0.25/kWh which is about 20% cheaper than that with a diesel generator as backup (\$0.31/kWh). The capital cost and the total NPC of this system are about 30% and 18% lower, respectively. The estimated LCOE of using only diesel engine which is a common solution in remote areas is \$0.56/kWh which is 55% and 45% more expensive in comparison with the hybrid system with biogas engine and diesel engine as backup, respectively. This means that regarding the economic, using the biogas engine as backup is more feasible than using the diesel engine in the hybrid scheme and also using only diesel engine generator.

Regarding CO₂ emission, 17 tons of CO₂ per year can be avoided by using biogas engine as backup in the optimized hybrid configuration compared to using the diesel engine as backup. If compared with the diesel based power generator, this system can save 48 tons of CO₂ during one year operation.

Due to the uncertainties related to the diesel fuel price, the capital cost and the O&M cost of the biogas engine generator set including the digester, a sensitivity analysis has been performed considering feasible possibilities. The sensitivity result shows under which circumstances a system including a biogas engine generator set is economically a better choice than one including a diesel engine generator set as backup.

This study focused on off-grid applications. Comparing electricity price, the grid electricity is significantly cheaper than when produced by isolated off-grid systems. However, taking into account different factors, i.e., number of clients in a remote village, distance between grid and community, long-term planning and high capital investment for grid extension, a micro-grid powered by the proposed stand-alone renewable energy system offers a more viable solution. Taking into account.

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